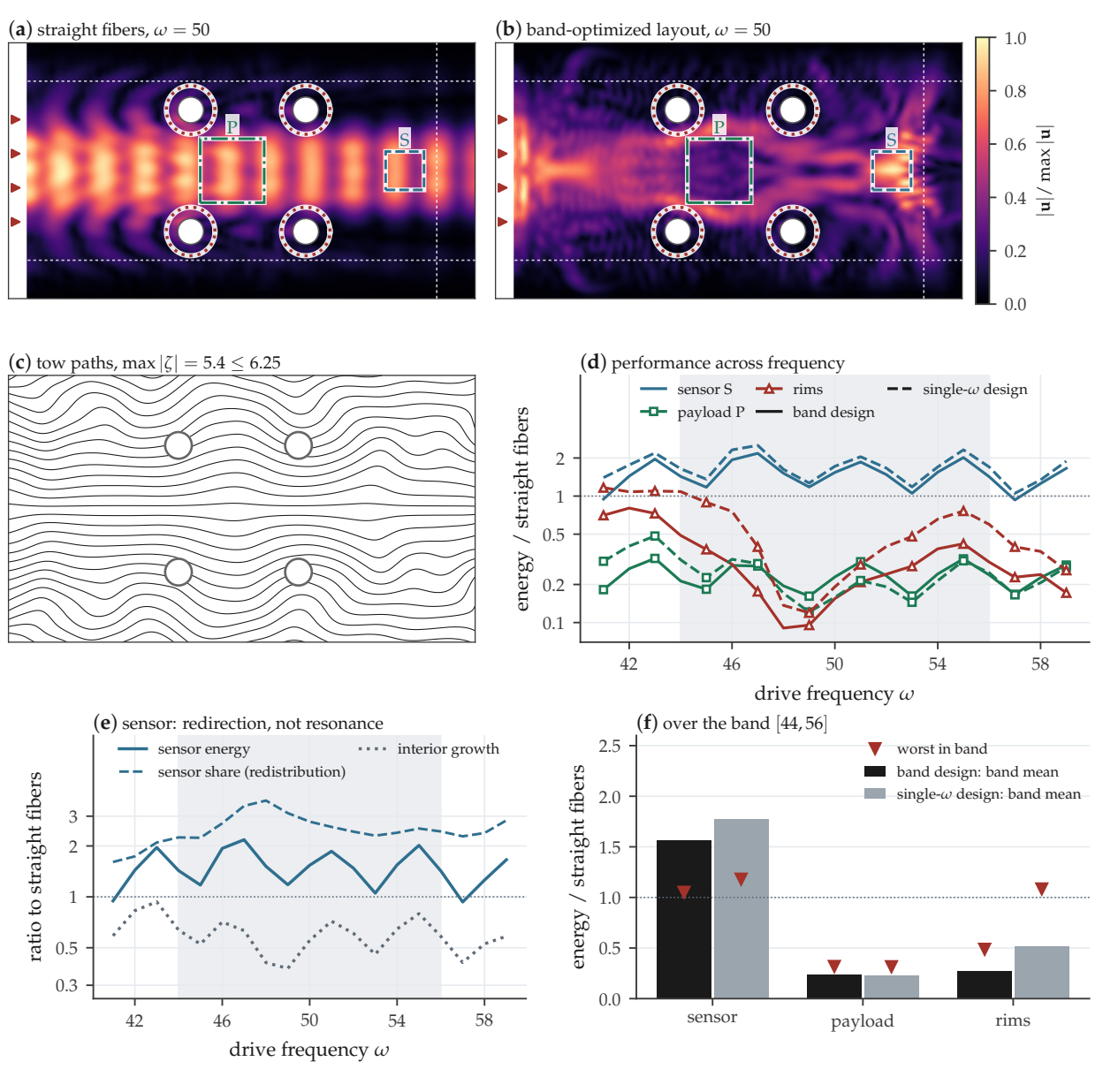

(a) straight fibers, ω = 50
(b) band-optimized layout, ω = 50
P
S
1.0
0.8
0.6
0.4
0.2
0.0
|u| / max |u|
(c) tow paths, max |ζ| = 5.4 ≤ 6.25
(d) performance across frequency
sensor S
payload P
rims
band design
single-ω design
energy / straight fibers
2
1
0.5
0.2
0.1
42
46
50
54
58
drive frequency ω
(e) sensor: redirection, not resonance
sensor energy
sensor share (redistribution)
interior growth
ratio to straight fibers
3
2
1
0.5
0.3
42
46
50
54
58
drive frequency ω
(f) over the band [44, 56]
worst in band
band design: band mean
single-ω design: band mean
energy / straight fibers
2.5
2.0
1.5
1.0
0.5
0.0
sensor
payload
rims